



OPEN Efficient self-healing framework for smart distribution networks

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This work presents a self-healing system designed for smart distribution networks, integrating fault detection, isolation, and power restoration with optimal network reconfiguration. Leveraging the Backward/Forward Sweep method for load flow analysis. The Particle Swarm Optimization algorithm for distributed generation sizing and location is used to improve the voltage profile for buses. The proposed design is validated on the IEEE 33-bus radial distribution system using MATLAB/Simulink and CYME software. Simulation results demonstrate a swift service restoration time, a reduction in power losses post-fault with two distributed generators, and an improved minimum voltage. This work enhances network reliability for the customers, voltage stability, and operational efficiency, offering a feasible and practical solution for investment in modern smart grids.

Keywords Self-healing, Smart grid, Distribution network, Distributed generation, PSO, IEEE 33-bus system, Fault isolation, Power restoration

The increasing complexity of modern electricity distribution systems, coupled with rising energy demands and environmental concerns, has necessitated a paradigm shift from traditional power grids to smart grids. Smart grids leverage advanced technologies to enhance the safety, reliability, and efficiency of power delivery, integrating distributed generation (DG), renewable energy sources, and real-time control mechanism^{1,2}. A key feature of smart grids is their ability to support two-way power flow, enabling distributed energy resources to complement centralized power generation. However, this evolution introduces new challenges, particularly in managing bidirectional power flows and ensuring system stability under fault conditions³. Traditional protection and control strategies designed for unidirectional power flow in radial networks are often inadequate for these dynamic systems, necessitating innovative approaches to fault management and service restoration.

Self-healing which is a cornerstone of smart grid technology, refers to the capability of a distribution network to autonomously detect, isolate, and recover from faults, thereby minimizing outages and enhancing system dependability^{4,5}. This concept has gained significant attention in light of the increasing pressures facing energy systems around the world due to climate change, aging infrastructure, and the integration of renewable energy sources⁶. Self-healing systems use advanced sensing, communications, and control technologies to isolate faulty components, reconfigure the grid, and restore power to healthy areas with minimal human intervention⁷.

Figure 1 illustrates a self-healing protection system that uses measuring devices and sensors to deliver electrical power to customers efficiently while maintaining high quality. These sensors are capable of detecting signals that indicate faults in the power lines of the network. The sensors form clusters and establish direct links to the main controller via a wireless connection. Multiple clusters communicate wirelessly with the main controller, enabling smooth grid operation. In the event of a fault, the system responds quickly to restore affected branches, ensuring a rapid recovery process.

Recent studies have highlighted the potential of self-healing to reduce outage durations and improve power quality, particularly in active distribution networks with embedded DGs^{8,9}. For instance, Torres et al.¹⁰ proposed a distributed intelligence system for self-healing, leveraging localized switch communication to handle complex fault scenarios, validated on real-world networks, demonstrating its efficacy in large-scale networks, while Shirazi et al.³ introduced an agent-based framework to optimize service restoration. Dong Li Jia et al.¹¹ proposed a layered architecture (base, support, application) and defined five operational states: normal, warning, critical, emergency, and recovery. This structure enhances outage management by enabling proactive fault mitigation. Elmitwally et al.¹² introduced a Multi-Agent Control System (MACS) to address voltage violations, feeder congestion, and reactive power coordination, demonstrating its efficacy in smart grids. Xu et al.¹³ developed a greedy algorithm to optimize remote-controlled switch placement, improving restoration capability with near-optimal solutions for large-scale networks. Zadsar et al.¹⁴ presented a two-layer algorithm integrating microgrids and direct load control, minimizing switching costs and load shedding while enhancing reliability. Zhao et al.¹⁵ introduces a specialized optimization model through case studies on modified IEEE 33-node and 123-node

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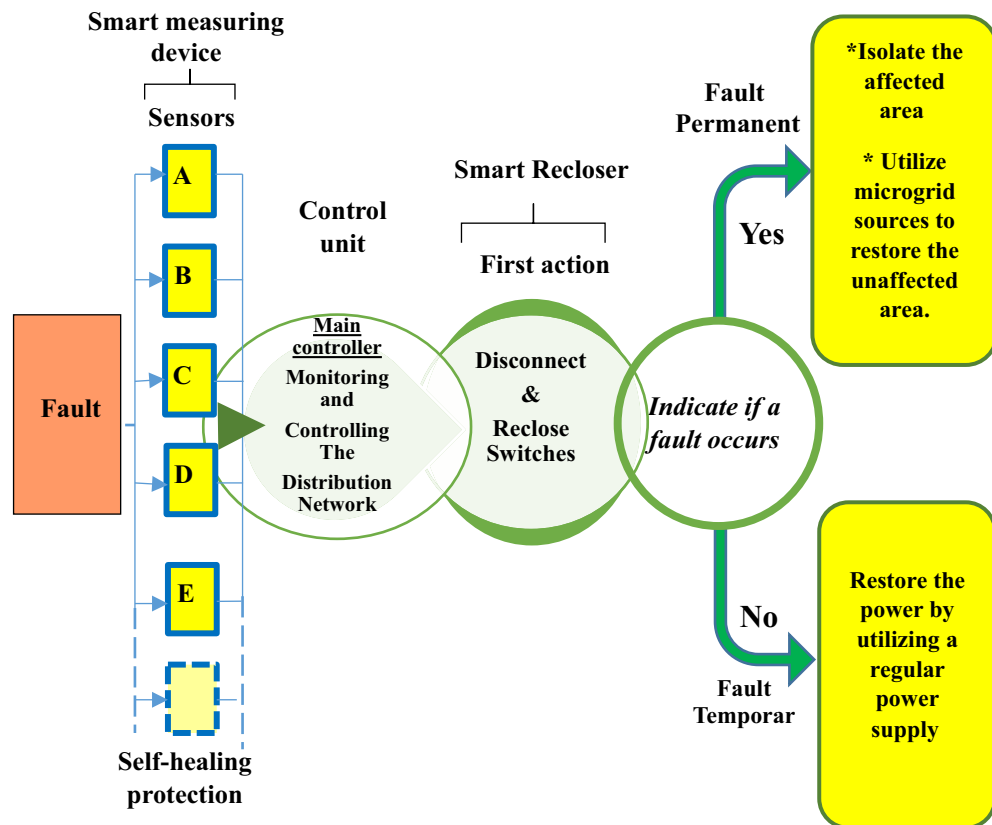


Fig. 1. Block diagram for a self-healing protection system structure.

networks. A binary hybrid algorithm is proposed to optimize switch states and DG operation modes, achieving minimal load curtailment, network losses, and voltage deviations, with demonstrated superior accuracy compared to BQPSO and BFOA solvers. The method effectively restores faulty ADNs to normal operation with minimal power losses.

Despite these advancements, significant research gaps remain. Many existing self-healing strategies focus on theoretical frameworks or small-scale simulations, with limited emphasis on practical implementation in real-world radial distribution systems^{16,17}. Furthermore, the integration of DGs into self-healing processes introduces additional complexity, as their optimal placement and sizing must be carefully considered to maximize restoration efficiency and minimize power losses^{11,13,18}. Generally, among the methods available, Particle Swarm Optimization (PSO) performs well and converges faster than alternatives like Genetic Algorithm (GA) and Whale Optimization Algorithm (WOA)¹⁹.

Studies show that PSO, GA, and WOA all reduce voltage deviations to below 0.05 p.u. across buses and significantly reduce power losses. However, PSO stands out for its speed, due to its swarm-based approach that efficiently shares information between particles. The GA, on the other hand, takes longer because it depends on evolutionary processes like selection, crossover, and mutation. While slower, GA is more robust and better suited for complex problems where thoroughness matters more than speed. The WOA offers similar performance to PSO, balancing exploration and exploitation, but its convergence time is sensitive to parameter tuning. Newer metaheuristics, such as Grey Wolf Optimization (GWO), show comparable results but are less studied in this context.

For the IEEE 33-bus system, PSO is one of the better optimization strategies used to resolve power system problems by increasing or decreasing the number of objective functions^{20,21}. This technique is used to optimize the number and location of the required DGs in the IEEE-33 bus system in this research, as a MATLAB code²². The economic and operational feasibility of such systems, particularly under varying fault scenarios and load conditions, also warrants further investigation²³.

Against this backdrop, this study proposes a novel self-healing methodology for smart distribution networks, utilizing the IEEE 33-bus test system as a benchmark. By combining the Backward/Forward Sweep load flow method with PSO for DG placement, this research aims to achieve rapid fault isolation, optimal network reconfiguration, and maximum power restoration while adhering to operational constraints. The proposed approach advances the state-of-the-art by offering a computationally efficient and scalable solution, validated through simulations in MATLAB and CYME software, with implications for enhancing the resilience of modern power distribution systems.

Methodology

IEEE-33 bus system model

The simulation of a power distribution network was performed using MATLAB R2017b and CYME 9.0, with the IEEE 33-bus test system adopted as the reference topology. The CYME-based simulation entailed a systematic process, including the use of an equipment library, integration of load data with precise geospatial assignments, and specification of all components essential for power flow analysis, including line and load specifications, unbalanced load conditions, and fault calculations, thereby enhancing the accuracy of network simulations. It conducts load flow and power loss analyses using various sweep methods to determine voltage profiles and current magnitudes across the IEEE 33-bus system under diverse operating conditions. Furthermore, it supports the validation of restoration strategies through reliability metrics and provides limited real-time simulation capabilities. The load and line data for the IEEE 33-bus system are listed in Table 1²⁴.

CYME offers improved modeling accuracy and validation capabilities through its specialized distribution modeling features and adherence to standard performance metrics, thereby complementing the study's PSO-based optimization approach and self-healing system design and simulation utilizing MATLAB/Simulink.

Load flow analysis of the initial configuration revealed that the feeder was operating under abnormal conditions, with 19 sections experiencing under-voltage issues.

The research methodology was structured into three distinct phases: First, a load flow analysis was conducted to determine the voltage profiles and current magnitudes across the IEEE 33-bus system under steady-state conditions. The second phase focused on optimizing the sizing and placement of distributed generators (DGs) within the same test system to eliminate the under-voltage issues, employing the Particle Swarm Optimization (PSO) algorithm implemented in MATLAB. Finally, the third phase investigated the implementation of a self-healing protection scheme within a smart grid framework, applied to the IEEE 33-bus system both with and without DGs, leveraging the combined capabilities of MATLAB R2017b and CYME 9.0 to assess system performance and resilience.

Branch no.	From bus	To bus	R (Ω)	X (Ω)	Receiving bus	
					P (kW)	Q (kVAR)
1	1	2	0.0922	0.0470	0	0
2	2	3	0.4930	0.2511	100	60
3	3	4	0.3660	0.1864	90	40
4	4	5	0.3811	0.1941	120	80
5	5	6	0.8190	0.7070	60	30
6	6	7	0.1872	0.6188	60	20
7	7	8	0.7114	0.2351	60	20
8	8	9	1.0300	0.7400	200	100
9	9	10	1.0440	0.7400	200	100
10	10	11	0.1966	0.0650	60	20
11	11	12	0.3744	0.1238	45	30
12	12	13	1.4680	1.1550	60	35
13	13	14	0.5416	0.7129	60	35
14	14	15	0.5910	0.5260	120	80
15	15	16	0.7463	0.5450	60	10
16	16	17	1.2890	1.7210	60	20
17	17	18	0.7320	0.5740	60	20
18	18	19	0.1640	0.1565	90	40
19	19	20	1.5042	1.3554	90	40
20	20	21	0.4095	0.4784	90	40
21	21	22	0.7089	0.9373	90	40
22	22	23	0.4512	0.3083	90	50
23	23	24	0.8980	0.7091	420	200
24	24	25	0.8960	0.7011	420	200
25	25	26	0.2030	0.1034	60	25
26	26	27	0.2842	0.1447	60	25
27	27	28	1.0590	0.9337	60	20
28	28	29	0.8042	0.7006	120	70
29	29	30	0.5075	0.2585	200	600
30	30	31	0.9744	0.9630	150	70
31	31	32	0.3105	0.3619	210	100
32	32	33	0.3410	0.5302	90	40

Table 1. The line and load data for the 33-bus test distribution system²⁴.

The IEEE 33-bus radial distribution system, operating at 12.66 kV, serves as the testbed²⁵. It comprises 33 buses, 32 sectional switches (Sw-1 to Sw-32, normally closed), and 5 tie switches (Sw-33 to Sw-37, normally open), with a total load of 3715 kW active and 2300 kVAR reactive power. The feeder's current capacity is 208.6 A, with initial power losses of 202.66 kW and a minimum voltage of 0.91 p.u. at bus 17²⁶. Figure 2 depicts the single-line diagram of the system, illustrating its radial topology and switch locations.

Load flow analysis

Load flow analysis for the IEEE 33-bus distribution system can be conducted using either MATLAB-based codes or CYME software. To assess result accuracy and performance, this study investigates three methods: the Backward/Forward Sweep, Newton-Raphson, and Gauss-Seidel techniques.

The Newton-Raphson method is known for its fast (quadratic) convergence and high accuracy, but it involves significant computational effort due to the repeated calculation and inversion of the Jacobian matrix. The Gauss-Seidel method, while simpler and more memory-efficient, is better suited for smaller systems due to its slower convergence rate²⁷. Both methods, however, face limitations when applied to radial distribution systems, primarily due to high R/X ratios and convergence difficulties.

In contrast, CYME software offers a more efficient and practical solution for radial systems. It includes specialized load flow analysis tools designed to accommodate the specific characteristics of distribution networks, providing accurate results with reduced computational complexity.

The BW/FW method, ideal for radial systems with high R/X ratios²⁸ iteratively solves load flow equations:

- Backward Sweep: Calculates branch currents using Kirchhoff's Current Law:

$$I_i = \left(\frac{S_i}{V_i} \right)^* \quad (1)$$

- Forward Sweep: Updates voltages via Kirchhoff's Voltage Law:

$$V_j = V_i + Z_{j,i} \times I_{j,i} \quad (2)$$

where S_i is the complex power, V_i is the voltage at the bus i , and $Z_{j,i}$ is the impedance between buses j and i .

- The calculated voltage error, which represents the voltage difference between bus 1 voltage (V_1) and the main source voltage (V_s), is.

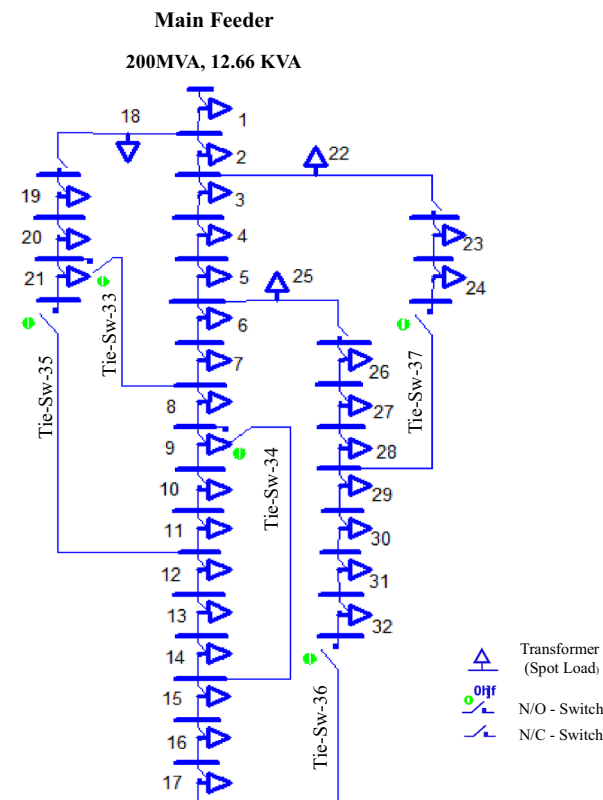


Fig. 2. IEEE 33-bus radial distribution test system single-line diagram.

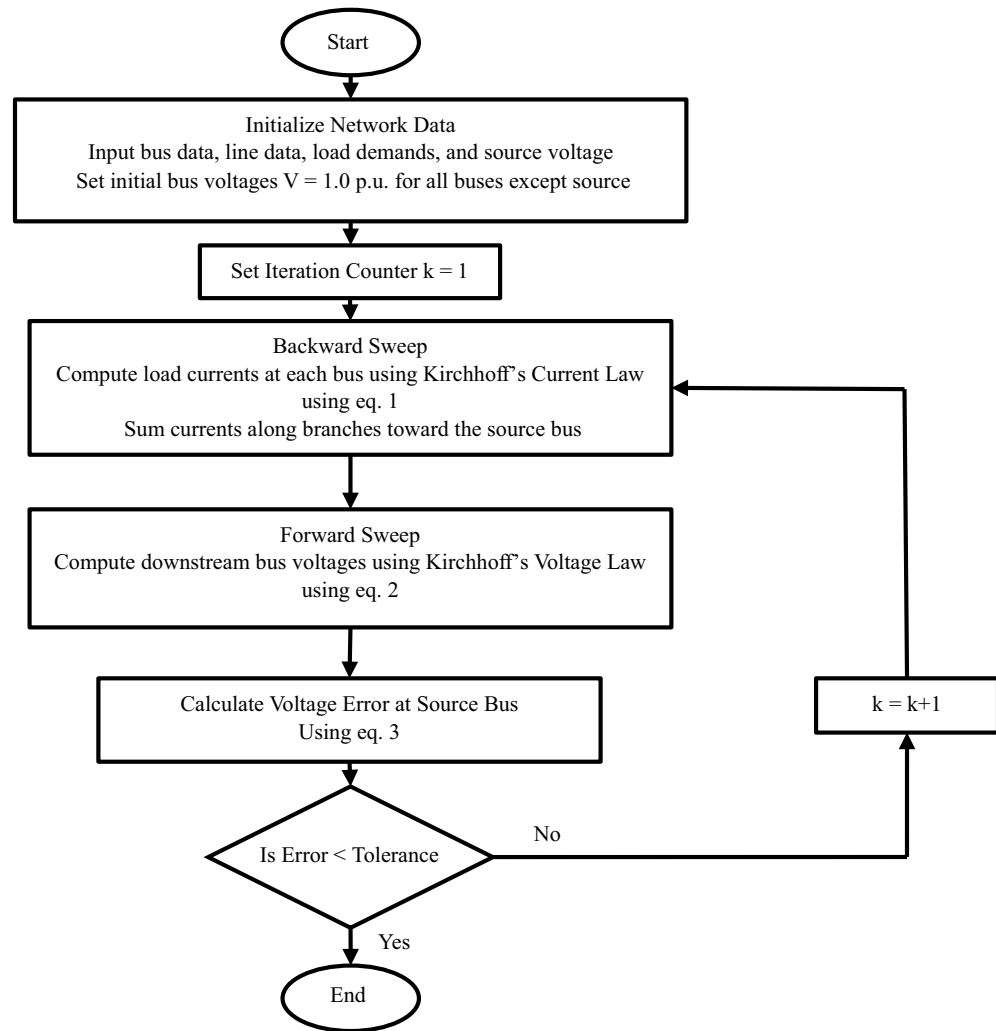


Fig. 3. Backward/forward sweep load flow algorithm for radial distribution systems.

Loading method	Current/phase (amp)	Total load (kW)	Total load (kVAR)	Power losses (kW)	Min. bus voltage (p.u)	Min. voltage location
Newton-Raphson	208.64	3715	2240	202.66	0.90	Bus 17
Gauss-Seidel	208	3715	2240	201.95	0.9138	Bus 17
Backward/forward	208.6	3715	2240	202.4	0.91	Bus 17

Table 2. The IEEE 33-bus load-flow results.

$$Error = ||V_s| - |V_1|| \quad (3)$$

The flowchart illustrating the Backward/Forward Sweep load flow algorithm for the IEEE 33-bus distribution network is illustrated in Fig. 3.

Results in Table 2, align with the traditional methods of Newton-Raphson and Gauss-Seidel methods, confirming robustness.

Optimization utilizing DG

PSO optimizes DG placement and sizing to minimize total power losses:

$$PL = \sum_{i=1}^N \sum_{j=1}^N [\alpha_{ij} (P_i P_j + Q_i Q_j) + \beta_{ij} (Q_i Q_j + P_i P_j)] \quad (4)$$

where $\alpha_{ij} = \frac{r_{ij}}{v_i v_j} \cos(\delta_i - \delta_j)$, $\beta_{ij} = \frac{r_{ij}}{v_i v_j} \sin(\delta_i - \delta_j)$, $Z_{ij} = r_{ij} + jx_{ij}$ are the ij^{th} element of [Z bus] matrix, $P_i = P_{Gi} - P_{Di}$ is the active power of the buses, $Q_i = Q_{Gi} - Q_{Di}$ is the reactive power of the

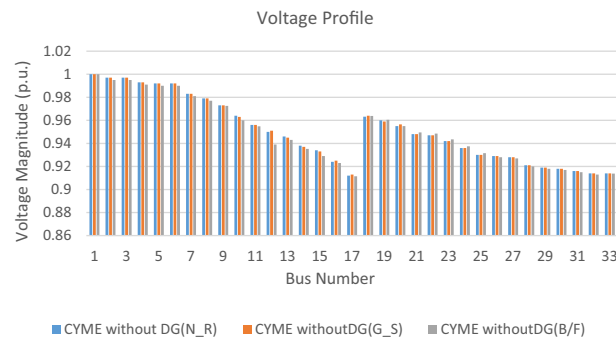


Fig. 4. Voltage profile (normal operation).

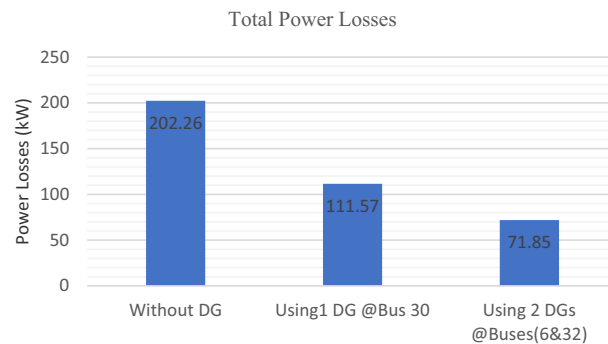


Fig. 5. The total power losses in the IEEE 33-bus distribution system.

buses, P_{Gi} & Q_{Gi} are the generator's active and reactive power injected into the busbar, P_{Di} & Q_{Di} are the demand's (loads) active and reactive power.

The bus voltage levels must not exceed the limitations, where V_{min} and V_{max} are the lowest and maximum r.m.s voltages on a specific bus. The permitted limits are as follows^{29,30}:

$$V_{min} = 0.95 \text{ p.u. and } V_{max} = 1.05 \text{ p.u.}$$

The current flowing through the load line in the grid has allowable limits and represents another constraint¹³.

$$I_j \leq I_{jmax} \quad (5)$$

By leveraging PSO-based DG placement and dynamic network reconfiguration, the system maintains voltage stability, minimizes losses, and ensures reliable service restoration under varying load and generation conditions. The adaptive control strategy is capable of responding to the unpredictable nature of power flow directions by dynamically adjusting switch states and load paths. Furthermore, the consideration of DG output in the reconfiguration logic enhances the system's ability to manage reverse or bidirectional power flow, prevent undervoltage and overvoltage conditions, and maintain balanced feeder loading.

Load flow calculation results

Normal operation

The bus voltages of the system in its original configuration during normal operation were calculated using the Backward/Forward Sweep load flow method, as shown in Fig. 4. The results, computed in MATLAB R2017b and CYME and presented in Table 2, align closely with those obtained from the Newton-Raphson and Gauss-Seidel methods. Figure 4 illustrates the voltage profile of the IEEE 33-bus test system derived from the Backward/Forward Sweep method, alongside a comparison with results from other methods. Initial load flow analysis revealed under-voltage in 19 sections (< 0.95 p.u.), with a minimum of 0.90 p.u. at bus 17 and losses of 202.66 kW.

With DG placement

To minimize active power losses, the proposed method employs the Particle Swarm Optimization (PSO) technique to determine the optimal number, sizes, and locations of distributed generators (DGs). Figure 5 visually compares power losses across different DG deployment scenarios, indicating that using two DGs is the most effective choice. The PSO algorithm identifies bus bars 6 and 32 as the optimal locations for these DGs.

Load flow analysis conducted after distributed generation (DG) placement revealed that section 6 requires 1310.805 kVA per phase, and section 32 requires 873.8701 kVA per phase. Table 3 summarizes the load conditions before and after reactive power compensation. Initially, the system exhibited a minimum voltage of

Condition	Losses (kW)	Min. voltage (p.u.)	Current (A)
Without DG	202.66	0.90 (bus 17)	208.4
With 2 DGs	71.85	0.97 (bus 17)	74

Table 3. System performance with/without DGs.

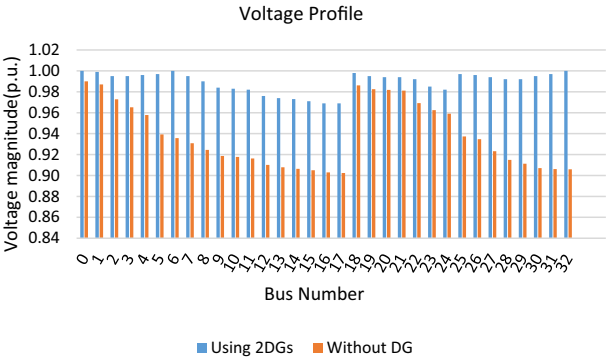


Fig. 6. Voltage profile with/without DGs.

0.90 per unit, as confirmed by load flow analysis. Following optimal DG placement, the system losses decreased to 71.85 kW, with the minimum voltage reaching 0.97 per unit at bus 17. Figure 6 illustrates the enhanced voltage profile, highlighting DG benefits³⁰.

Self-healing framework

In this work, to apply the self-healing methodology on the IEEE 33-bus test system using CYME software, the fault was chosen to be simulated in Sect. 8 because of the ability of fault isolation and multiple choices to reconfigure the network and restore power to this area using fewer numbers of switches. This study investigates a three-phase short-circuit fault introduced in the transmission line between bus-bar 8 and bus-bar 9 of the IEEE 33-bus distribution network. This type of fault is classified as a big disturbance due to its substantial impact on voltage profiles, power flow, and overall system stability. In contrast to small disturbances, such as small load variations or single-phase faults, a three-phase fault necessitates complex network reconfiguration. The proposed self-healing approach proves the ability to manage such a fault effectively.

As soon as a fault occurs on the line between bus bar 8 and bus bar 9, the protection relay located at bus 8 opens the switch (Sw-8) to isolate the rest of the line to prevent the fault from affecting the healthy lines located before the fault location, thereby avoiding a shutdown of the entire power distribution network. As a consequence, the power to Sects. 8 to 17 is disrupted, as depicted in Fig. 7. The self-healing protection system responds by opening the switch (Sw-9) to isolate and clear the faulty line completely.

The self-healing process unfolds in three stages:

- Fault Detection and Isolation: Sensors monitor voltage and detect anomalies (e.g., a fault in Sect. 8). The control unit triggers the relay (Sw-8 opens) to isolate the faulty line, interrupting 815 kW across Sects. 8–17²³.
- Network Reconfiguration: Optimal switching (open Sw-9, close Sw-35) minimizes losses and balances load, maintaining radiality³¹.
- Power Restoration: Tie switches (Sw-35) close to reroute power, restoring service total, leveraging automated control signals¹⁰.

To achieve optimal network reconfiguration and power restoration, voltage drop areas were addressed by strategically deploying distributed generators (DGs) with calculated power ratings and locations. Figure 8 illustrates the optimal switching plan for the IEEE 33-bus system, which restores power to feeders 9 through 17 via switch Sw-35 after isolating the fault in Sect. 8 using DGs.

The self-healing process for the power distribution network is summarized in a detailed flowchart, presented in Fig. 9, which systematically outlines the sequential steps involved in the self-healing mechanism.

Simulink modelling and simulation setup
Post-fault self-healing

Case (A): application of self-healing methodology on the IEEE 33-bus test system using CYME software
The self-healing methodology was applied to the IEEE 33-bus test system using CYME software, with a fault simulated in Sect. 8 due to its suitability for fault isolation and multiple reconfiguration options to restore power with minimal switch operations. The self-healing protection system responds by opening switches Sw-8 and Sw-9 to isolate the fault, resulting in a power disruption to Sects. 8 through 17, as shown in Fig. 7. This outage affects a total unserved consumer load of 815 kW.

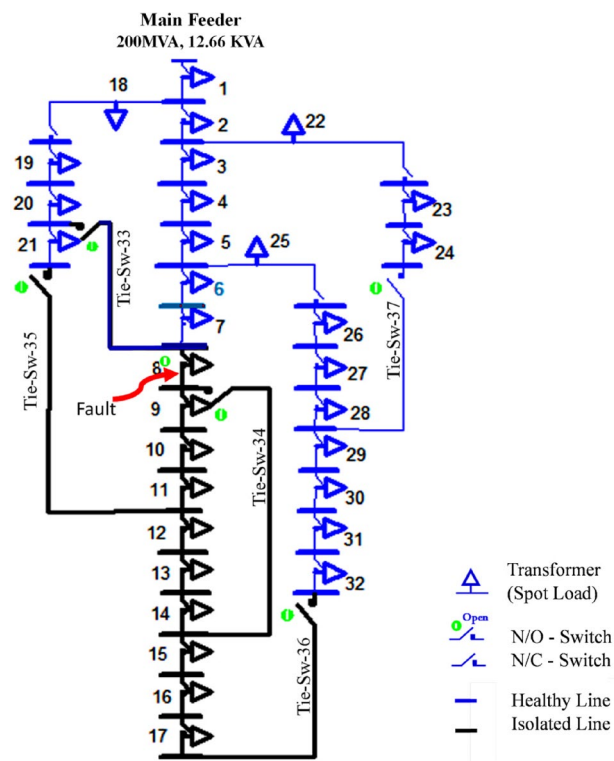


Fig. 7. The IEEE 33-bus system after isolating the fault in section 8.

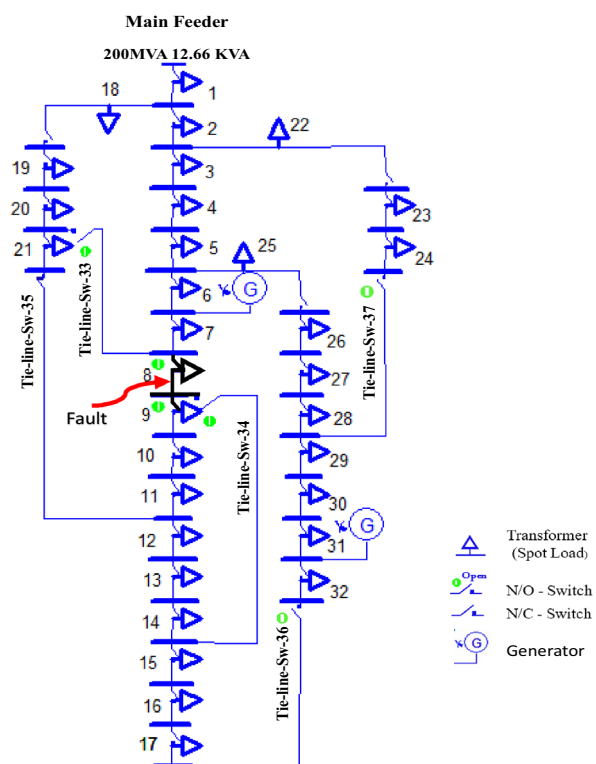


Fig. 8. The IEEE 33-bus system optimal switching plan to restore power to feeders 9 to 17 through Sw-35 after isolating the fault on Sect. 8 using DGs.

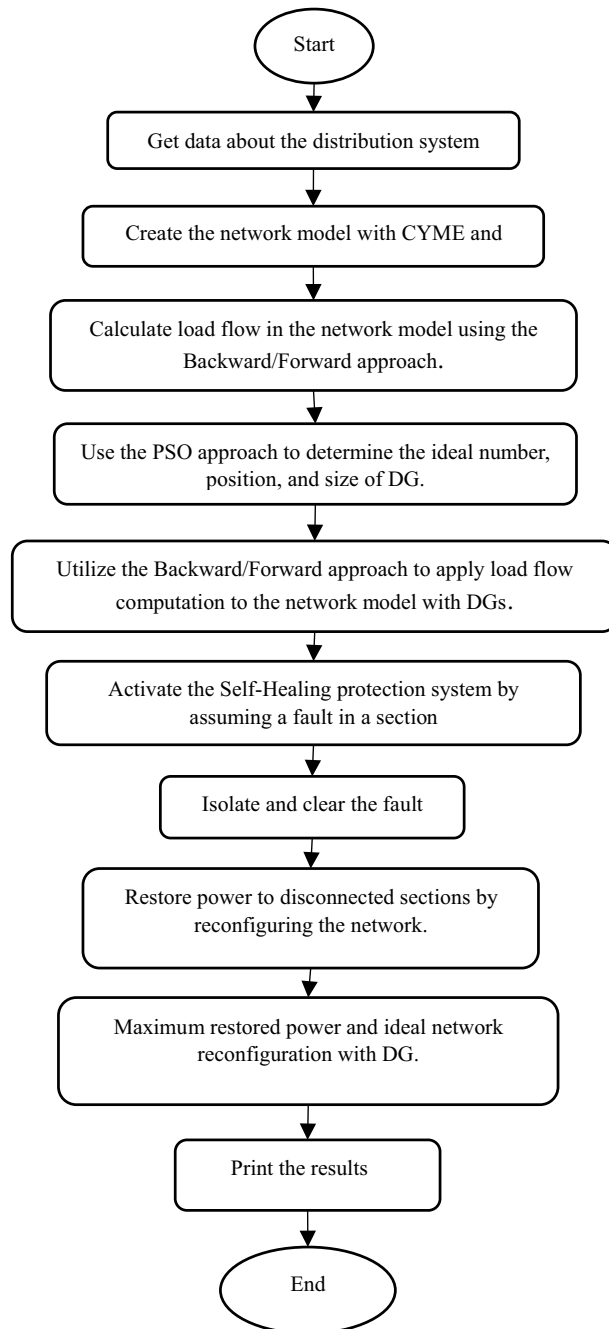


Fig. 9. Proposed methodology procedure flowchart.

Network reconfiguration was effectively carried out using self-healing techniques to strategically manage switch operations, ensuring optimal load transfer and power restoration for consumers affected by the outage in Sect. 8. The optimal setup for Sect. 8 involves opening switch Sw-9 and closing the normally open tie switch Sw-35, as outlined in Table 4, for scenarios both with and without distributed generation (DG). This configuration facilitates the transfer of 815 kW through the tie line switches after fault isolation. The resulting network layouts are illustrated in Fig. 8.

Case (B): application of the self-healing system using matlab/simulink

In this study, the simulation framework for evaluating self-healing mechanisms in power distribution networks is developed using MATLAB/Simulink, with the IEEE 33-bus test system serving as the foundational topology. This section delineates the configuration of the simulation environment, encompassing the hardware and software infrastructure, alongside a comprehensive description of the model components, including bus configurations, load profiles, power sources, and switching mechanisms.

Metric		Proposed method	
		Without DG	With 2DGs
Operating switches	Open switches	Sw-9	Sw-9
	Close switches	Tie-Sw-35	Tie-Sw-35
Number of operating switches		2	2
Total power losses (kW)		144.97	48.6
Min. voltage (p.u.) Service bus (before fault)		0.90	0.97
Location of min. voltage		bus 17	bus 17
Max. voltage (p.u.) service bus		1	1

Table 4. The optimal switching plan after isolating the fault at sect. 8.

Central to the simulation is the control unit model, precisely designed in MATLAB/Simulink to coordinate the operation of smart protection devices. The Simulink model represents the dynamic behavior of these devices, offering a graphical representation of their interactions within the distribution network. Key components include bus signal blocks, which emulate protective relays with configurable parameters such as overcurrent settings, time delays, and trip characteristics; circuit breaker blocks, which model the response to fault conditions through controlled opening and closing actions; and switch blocks, which simulate network reconfiguration by isolating faulty segments or redirecting power flow. The control unit block integrates decision-making algorithms, processing inputs from sensors and relays to execute coordinated actions, such as fault isolation or power restoration.

Additionally, a measurement unit block collects and processes real-time system data, providing critical inputs for the control unit's decision-making process. This model facilitates the simulation of diverse fault scenarios, enabling thorough analysis of device coordination, response times, and the overall efficacy of the smart protection system in enhancing the reliability and stability of the distribution network. Figure 10 shows the Simulink control unit, integrating relays, breakers, and switches.

The self-healing approach was implemented in MATLAB/Simulink using protective devices capable of receiving remote commands to respond rapidly to faults. In this scenario, a fault is simulated in Sect. 8, as shown in Fig. 11. When the fault occurs, the current exceeds its rated limit, prompting the monitoring unit to detect the abnormal current surge. This triggers a signal to the comparison unit, which evaluates the currents in each feeder segment against the relay settings. To isolate the fault, the comparison unit sends a signal to switch Sw-8, which opens within 0.02 s. Consequently, the power supply to Sects. 8 through 17 is disrupted, resulting in an unserved consumer load of 815 kW.

Figure 12 demonstrates that when implementing the self-healing process without distributed generators (DGs), the minimum voltage of 0.90 p.u. presented at bus 17, and the system losses calculated to be 144.97 kW. With the integration of DGs, the minimum voltage improved to 0.973 p.u. at bus 17, and the losses were substantially reduced to 48.6 kW as detailed in Table 4. Figure 12 presents the post-fault voltage profile, while Fig. 13 compares branch currents.

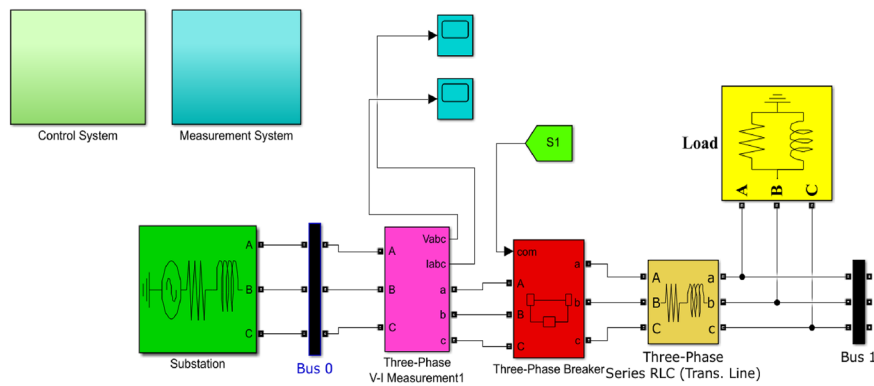
Figure 13 offers the voltage profile before/after activation of the fault event in the self-healing system. The pre-fault voltage at bus 9, steady at 0.984 p.u., demonstrates the enhanced steady-state condition achieved through optimal DGs placement. Following the fault, the voltage stabilizes at 0.972 p.u. The slight voltage reduction at bus 9 is due to load redistribution and line impedance effects post-reconfiguration, yet it remains within the acceptable range of 0.95–1.05 p.u. Additionally, Fig. 13 indicates that the post-restoration minimum system voltage is 0.94 p.u. at bus 32. This robust dynamic response highlights the system's capability to maintain voltage stability during fault conditions, reinforcing its reliability as a key feature of a smart grid.

Figure 14 illustrates the branch current loading in the IEEE 33-bus system under two conditions: normal operation and post-fault isolation. During normal operation, the main feeder carries a load current of 106.2 A per phase. Following a fault in section 8 and the subsequent self-healing process with fault isolation, the total recovered load current across the 33-bus system reduces to 74.6 A per phase, as detailed in Table 5.

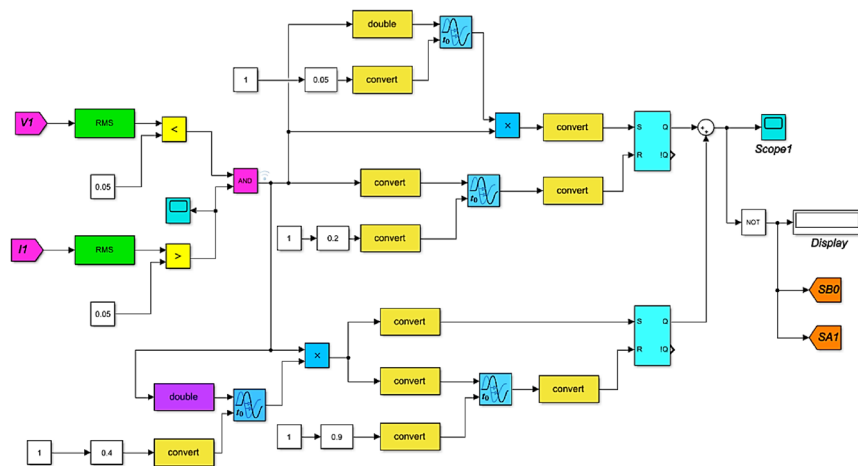
Figure 15 displays a time-domain plot of the voltage magnitude at bus 9 during a fault in Sect. 8, simulated using MATLAB/Simulink. The x-axis represents time in seconds, and the y-axis shows the voltage value in per unit (p.u.). Before the fault, the voltage was stable at approximately 0.98 p.u., reflecting the improved voltage profile due to DGs placed at buses 6 and 32.

At the fault onset ($t=0.1$ s.), the voltage drops sharply due to the disturbance. Within 0.02 s., the fault is detected and isolated by opening switch Sw-8, resulting in a transient voltage dip. The self-healing process then activates, opening switch Sw-9 and closing switch Sw-35 to restore power. At $t=0.16$ s, the voltage stabilizes at approximately 0.97 p.u., confirming successful power restoration. This rapid recovery, achieved within 60 ms, highlights the effectiveness of the control unit and DG support in mitigating voltage sags and ensuring system stability. During the fault event, the self-healing approach underscores the system's capability to maintain stability and ensure reliable power restoration. The self-healing system significantly reduced overall power losses by 76% (from 202.66 kW to 48.6 kW).

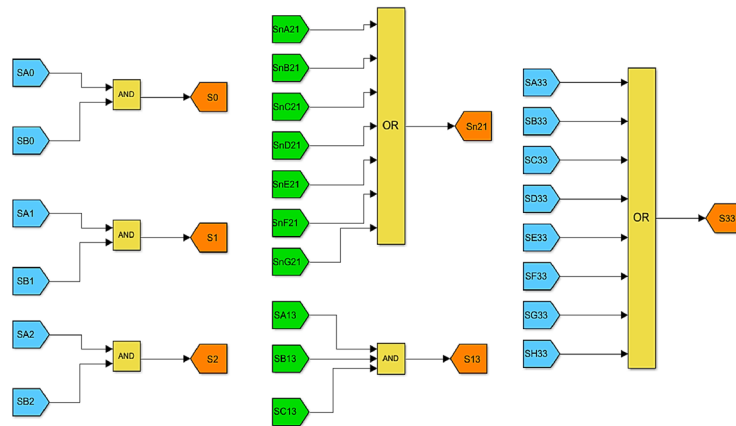
Although direct comparisons with other studies are limited by the lack of standardized benchmarks for the IEEE 33-bus system, Table 6 compares the quantitative performance of the proposed self-healing framework based on PSO for DG placement with related works in the literature. The comparison includes voltage-profile



(a) The Simulink model with integrating components, system units, relays, breakers, and switches.



(b) The bus current and voltage monitoring components in the measurement system unit.



(c) The logic unit model diagram in the control system.

Fig. 10. The configuration of the system components in the Simulink model. (a) The Simulink model with integrating components, system units, relays, breakers, and switches. (b) The bus current and voltage monitoring components in the measurement system unit. (c) The logic unit model diagram in the control system.

improvement, loss reduction, and service restoration time. Additionally, a comparison between the base case load conditions before and after optimal DG placement is presented in Table 3.

The proposed self-healing mechanism demonstrates robust performance under varying load conditions and fault scenarios, including multi-point faults and real-time load variations, through its adaptive control strategy

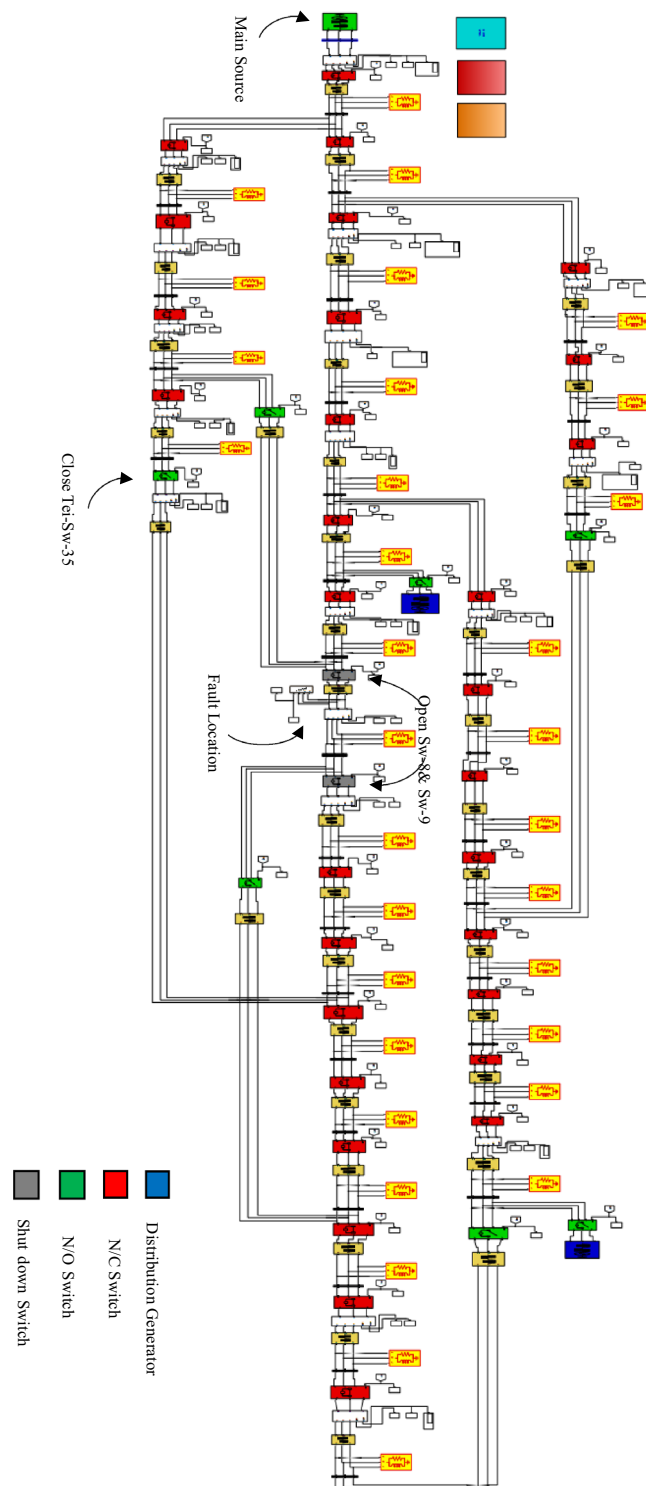


Fig. 11. The IEEE 33 bus test system is supported with 2 DGs Simulink mode.

and optimized network reconfiguration. It adapts to different load conditions through PSO-based DG placement and network reconfiguration, ensuring voltage stability and minimizing losses.

In fault scenarios, the system effectively handles single-point faults and is also capable of managing multi-point faults using the same control strategy. It utilizes automated switching and centralized control functions for monitoring, fault detection, isolation, and reconfiguration, all managed by a central control unit through fully controlled switches. Specifically, only three switches are used per fault: one bus-bar switch is tripped to isolate the

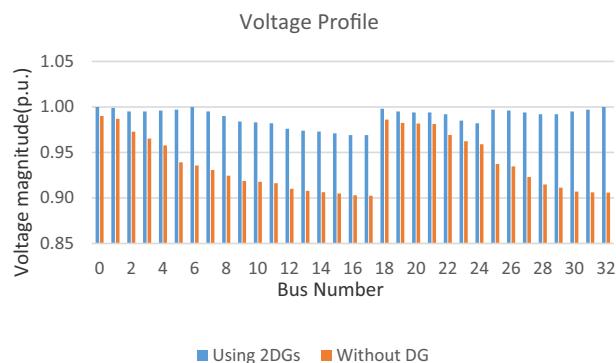


Fig. 12. The IEEE 33-bus system voltage profile with and without 2 DGs in MATLAB.

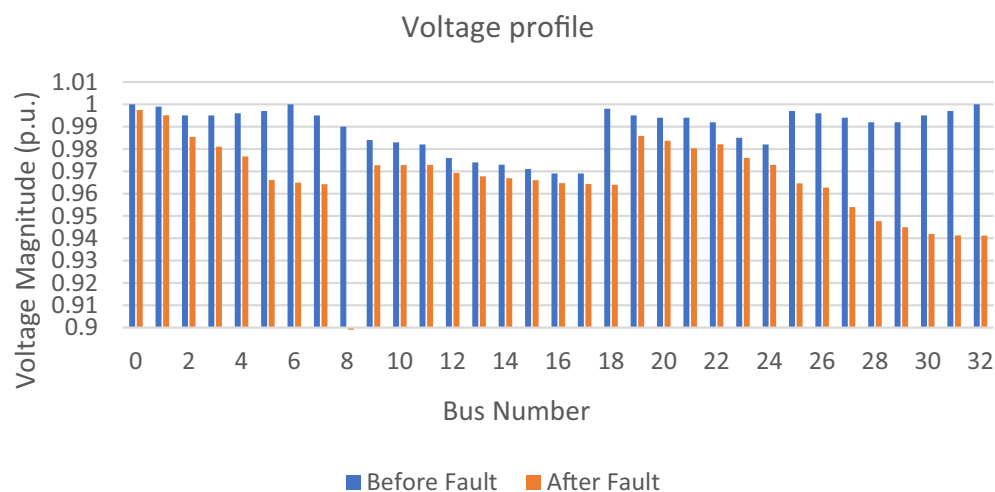


Fig. 13. Voltage profile before/after fault isolation with DG.

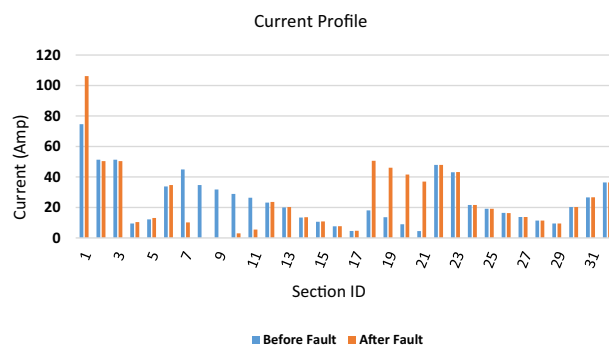


Fig. 14. Branch currents before/after fault isolation.

faulty section based on a protection relay signal; the next downstream bus-bar switch is automatically tripped to isolate the faulty line; and finally, the tie switch is closed automatically to restore service to the healthy sections.

Conclusion

This work developed a MATLAB/Simulink-based simulation model for self-healing in power distribution networks, using the IEEE 33-bus test system as a test case. The simulation model is designed to integrate smart protection relays, circuit breakers, and switches, which are coordinated by a control unit for rapid fault response. A fault assumed to occur in Sect. 8 was detected by smart measuring sensors and isolated within 0.02 s by opening switch Sw-8, resulting in an outage of 815 kW of load across Sects. 8–17. The proposed self-healing process

Description	Before fault	After service restoration
Max. voltage (p. u.)	1	1
Min. voltage (p. u.)	0. 97 (bus 17)	0. 94 (bus 32)
Power losses (kW)	71. 8	48. 6
Current/phase (Amp)	106. 2	74. 6

Table 5. Simulation results before and after fault isolation and service restoration using 2DGs.

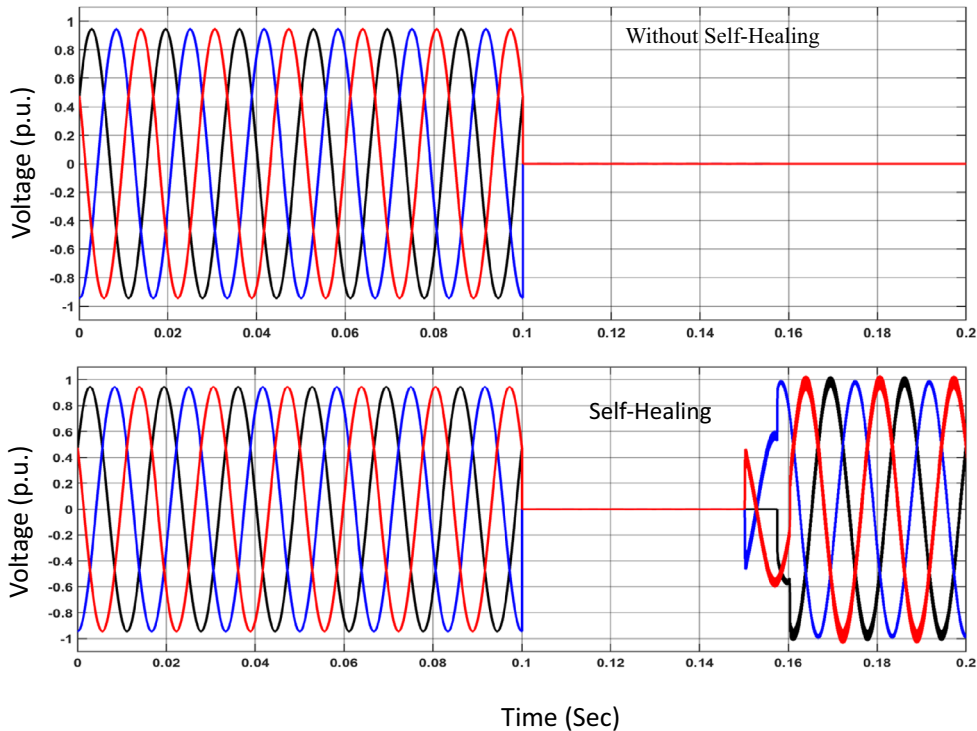


Fig. 15. Voltage on bus-bar 9 before and after the application of the self-healing process, using DGs, during a fault occurrence.

References	Method	Voltage profile improvement value (p.u.)	Total power loss (kW)	Loss reduction (kW)	Service restoration time (seconds)
32	FWA	0.0282	139.98	32.69	–
33	IGA	0.0786	11.59	191.08	–
	IPSO	0.0784	11.12	191.55	
	ECSO	0.0801	12.16	190.51	
	Firefly-DN	0.0243	142.80	139.88	
34	SPSO		140.42	140.48	–
	SBAT		141.20	140.36	–
35	SPSO	0.037	112.58	89.9	–
36	RCS	–	–	–	120 (2 min.)
This work	PSO	0.07	71.85	130.81	0.06

Table 6. Comparison of the proposed work with other works on a 33-bus system.

restored power by reconfiguring switches Sw-9 and Sw-35, the voltage at bus 9 stabilized to 0.947 p.u. within 60 milliseconds. The overall post-fault power losses reduction achieved from 202.66 kW to 48.6 kW (76%). The system demonstrates enhanced reliability and stability supported by distributed generators, as observed through Simulink's real-time data processing and measurements. The fast response and power recovery highlight the excellent performance of the control unit in eliminating voltage drops, reducing line outage time, and ensuring network resilience and reliability.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request by email (m.omran@urmia.ac.ir).

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Author contributions

F.M. worked on study research, wrote the main manuscript text, performed calculations, worked on CYME and MATLAB/Simulink software modelling, and results analysis; B.T. supervised all work and results; and M.O. wrote the MATLAB/Simulink codes and prepared figures. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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